

Context-Aware Nutritional Intelligence (CANI) Agent

“Like having a personal nutritionist beside you every day!”

1. Short Product Description

CANI (கனி) is a Context-Aware Nutritional Intelligence AI Agent designed to serve as a proactive nutrition and wellness companion. By integrating real-time weather data, environmental conditions, meal schedules, hydration requirements, and conversational AI, CANI provides personalized guidance that helps users maintain healthier daily habits.

Unlike traditional reminder applications that rely on fixed schedules, CANI adapts its recommendations based on context, including climate conditions, time of day, user interactions, and device activity. The agent operates seamlessly in the background and becomes active only when the user is actively engaged with their device.

2. Unique Value Proposition (USP)

Most wellness applications provide generic reminders that are identical for every user. CANI delivers context-aware recommendations by combining environmental intelligence, behavioral reminders, and conversational assistance into a single AI-driven system.

For example, instead of reminding users to "drink more water," CANI can estimate hydration needs based on local temperature, humidity, and time of day, then provide actionable guidance using intuitive measurements such as glasses of water.

3. Project Overview

CANI is an AI-powered wellness assistant that helps users maintain healthy hydration and eating habits throughout the day. The system continuously monitors relevant environmental factors and schedules personalized reminders for hydration, meals, and wellness activities.

The agent supports natural language conversations, allowing users to ask nutrition-related questions and receive region-specific, climate-appropriate recommendations. By combining proactive reminders with conversational intelligence, CANI aims to simulate the experience of having a personal nutrition companion available at all times..

4. Key Features

Feature	Context
Smart Device Awareness	• Detects system startup, shutdown, lock, unlock, and sleep states. • Activates only when the user is actively using the device. • Automatically enters a low-power inactive mode during system inactivity.
Climate-Aware Hydration Monitoring	• Retrieves real-time weather information based on the user's location. • Estimates hydration requirements using temperature, humidity, and time of day. • Provides hydration reminders using simple measurements such as glasses instead of litres. • Tracks hydration progress throughout the day.
Intelligent Meal Reminders	• Reminds users about breakfast, lunch, and dinner. • Tracks meal confirmations and follow-up responses. • Encourages consistent eating schedules and healthy habits.
AI Nutrition Assistant	• Answers nutrition-related questions through natural language conversations. • Suggests region-specific and climate-appropriate meal options. • Supports personalized recommendations based on user preferences and context.
Conversational Voice Assistant	• Supports voice-based interaction and spoken notifications. • Enables hands-free wellness assistance. • Provides proactive wellness updates throughout the day.

Feature	Context
Daily Wellness Insights	• Generates end-of-day summaries of hydration and meal adherence. • Highlights wellness achievements and missed goals. • Encourages long-term habit formation through continuous feedback.

5. Expected Outcomes

CANI aims to improve users' hydration consistency, meal regularity, and nutritional awareness through proactive, personalized guidance. By combining environmental intelligence, conversational AI, and adaptive reminders, the system seeks to create a more engaging and effective wellness experience than traditional reminder-based applications.